Kernel Density Estimation for Multiclass Quantification (Extended Abstract)

Alejandro Moreo[1] (✉), Pablo González[2], and Juan José del Coz[2]

[1] Istituto di Scienza e Tecnologie dell'Informazione, Consiglio Nazionale delle Ricerche, Pisa, Italy alejandro.moreo@isti.cnr.it

[2] Artificial Intelligence Center, University of Oviedo, Gijón, Asturias, Spain, gonzalezgpablo@uniovi.es, juanjo@uniovi.es

Abstract. Predicting the class distribution in a population is more valuable for several disciplines, such as the social sciences, epidemiology, or market research, than predicting the individual labels of its members; this supervised task is generally known as *quantification* (or *class prevalence estimation*) in the machine learning literature. This extended abstract revisits our recent work [6], where we show that the multiclass extension of one of the most important families of methods—the distribution-matching (DM) approaches—is flawed. We then propose an alternative method based on multivariate kernel density estimation that better captures inter-class relationships.

Keywords: Multiclass Quantification, Kernel Density Estimation

1 Introduction

Quantification is the supervised task of estimating class prevalence values in a sample of unlabelled data [4]. Unlike classification, which focuses on individual predictions, quantification addresses applications where aggregate-level information is paramount—such as social sciences, epidemiology, and market research.

While most work in quantification has focused on the binary setting, many real-world applications require multiclass quantification, where the goal is to estimate the distribution over several mutually exclusive classes. Extending binary quantification methods to the multiclass case is not always straightforward. A prime example is the class of distribution-matching (DM) methods, which estimate prevalences by finding the best-fitting mixture of class-conditional distributions. In the binary case, DM methods represent the posterior probabilities using histograms; the mixture parameter represents the sought prevalence of the test distribution; a visual motivation is depicted in Figure 1.

However, when extending this strategy to the multiclass case, DM needs to model *one histogram per class* [2]. As histograms are univariate models, the inter-class dependencies are irretrievably lost. We propose an alternative representation based on multivariate kernel density estimation (KDE) for modelling the per-class distributions of posterior probabilities, which lie in the probability

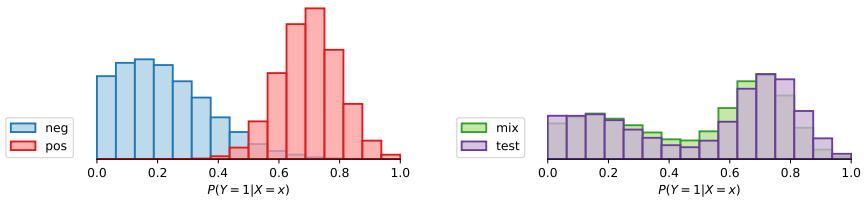

Fig. 1: DM in the binary case. Left: class-wise histograms of training posteriors. Right: the test distribution is approximated as a mixture of these histograms.

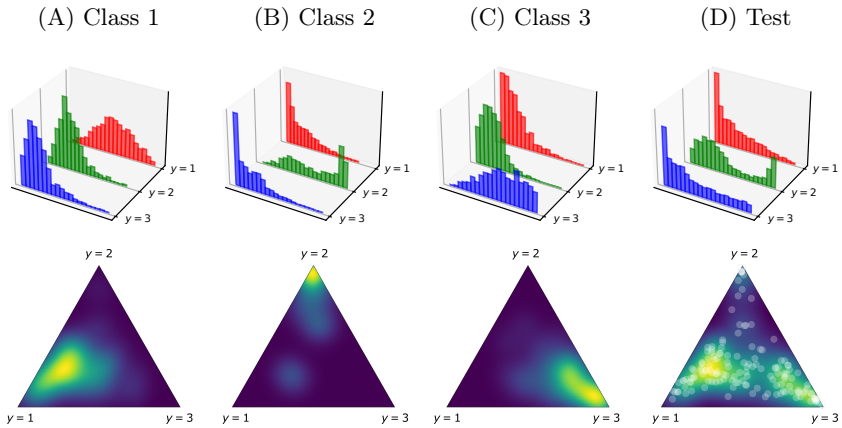

Fig. 2: Representation of a 3-class problem using class-wise histograms (top) and our KDE-based approach (bottom). Columns A–C show training class representations; column D shows the test set representation. The DM goal is to reconstruct D as a linear convex combination of A, B, and C.

simplex. Our recently published method KDEy [6] natively captures the joint structure of class probabilities; the intuition for a 3-class problem is depicted in Figure 2.

2 Quantification using Kernel Density Estimation

The DM problem for an n-class problem is framed as

$$(\hat{p}_1, \ldots, \hat{p}_n) = \underset{(p_1,\ldots,p_n)\in\Delta^{n-1}}{\arg\min} \mathcal{D}\left(\sum_{i=1}^{n} p_i M_i^{(tr)} \;||\; M^{(te)}\right) \tag{1}$$

where $\hat{p}_i$ is the class prevalence estimate for the ith class, $M_i^{(tr)}$ is a *model* for the training datapoints belonging to class i, $M^{(te)}$ is the model for the test datapoints, and $\mathcal{D}$ is any *divergence* function measuring the discrepancy between the mixture model and the test model.

Different mechanisms for representing sets have been explored in the quantification literature. A common approach is to represent individual instances using the posterior pobabilities estimated by a soft classifier h, and then aggregate them using a *permutation-invariant* function (e.g., the mean or histograms). We argue that representing the distribution of posteriors via kernel density estimation (KDE) is beneficial, as the model is multivariate and can thus capture inter-class dependencies that may exist in the data. Given a probabilistic classifier h and a kernel K with bandwidth λ (we use the Gaussian kernel) the KDE model for a set of datapoints $X = \{x_1, \ldots, x_m\}$ is given by: $M(x) = |X|^{-1} \sum_{x_i \in X} K\left[(h(x) - h(x_i))\lambda^{-1}\right]$. In our paper, we explore three variants, depending on the divergence measure being optimized in Equation 1:

- **KDEy-HD** adopts the popular Hellinger Distance (HD). As an example of an f-divergence, a closed-form expression for HD in continuous space is not known. We propose an efficient Monte Carlo approximation in the paper.
- **KDEy-CS** instead uses the Cauchy-Schwarz divergence, for which a closed-form solution is known. In the paper, we show an adaptation of this solution to the DM problem.
- **KDEy-ML** targets the Kullback-Leibler divergence, which, as we show in the paper, results in a maximum likelihood formulation.

3 KDEy in Action: Results and Insights

In our study, we compared the quantification performance of our three variants against 9 baseline quantification methods across 36 publicly available multiclass datasets of different nature, including textual and tabular data. For each dataset, we assessed the methods' ability to predict class prevalence on multiple test samples drawn from the original test pool, each exhibiting substantially different class distributions compared to the training set. In other words, we evaluated the quantifiers under conditions of *prior probability shift*—arguably the most relevant scenario for quantification—where class priors vary between training and deployment, while class-conditional distributions remain unchanged.

Performance was measured using the mean absolute error (MAE) and mean relative absolute error (MRAE) between the predicted and true class distributions, to enable a robust and interpretable comparison. Table 1 presents MRAE results averaged across groups of datasets, comparing our KDE-based variants (KDEy-HD, KDEy-CS, and KDEy-ML), which rely on multivariate density models, against their respective counterparts from the literature, which rely on univariate histogram-based methods (DM-HD, DM-CS) and a maximum likelihood approach (EMQ). The consistently lower errors achieved by our methods suggest that KDE-based representations are better suited for multiclass quantification. Figure 3 illustrates the sensitivity of the methods to their respective hyperparameters: the number of bins (*nbins*, shown in blue) for histogram-based models, and the kernel bandwidth (shown in orange) for KDE-based models, in terms of MAE: the KDE-based methods are generally more stable and easier to tune.

Data Group	**DM-HD**	**KDEy-HD**
Tweets (× 11)	0.422	0.405
UCI-ML Repo (× 24)	0.768	0.691
LeQua-T1B	0.911	0.783
Data Group	**DM-CS**	**KDEy-CS**
Tweets (× 11)	0.465	0.444
UCI-ML Repo (× 24)	0.841	0.749
LeQua-T1B	1.736	0.844
Data Group	**EMQ**	**KDEy-ML**
Tweets (× 11)	0.485	0.388
UCI-ML Repo (× 24)	0.751	0.685
LeQua-T1B	0.878	0.828

Table 1: Performance comparison of previous methods vs. proposed enhancements in terms of MRAE.

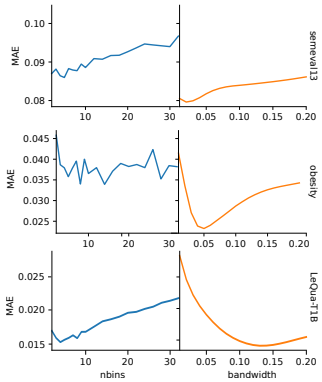

Fig. 3: Stability of methods based on hyperparameter tuning (exemplified in one dataset from each group).

The KDEy variants (particularly KDEy-ML) currently represent the state of the art in multiclass quantification, and are increasingly being adopted as robust solutions in a variety of application domains, including fairness monitoring in IR [5], graph-related applications [1], classifier accuracy prediction [8], sentiment quantification [7], and medical imaging for healthcare [3].

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